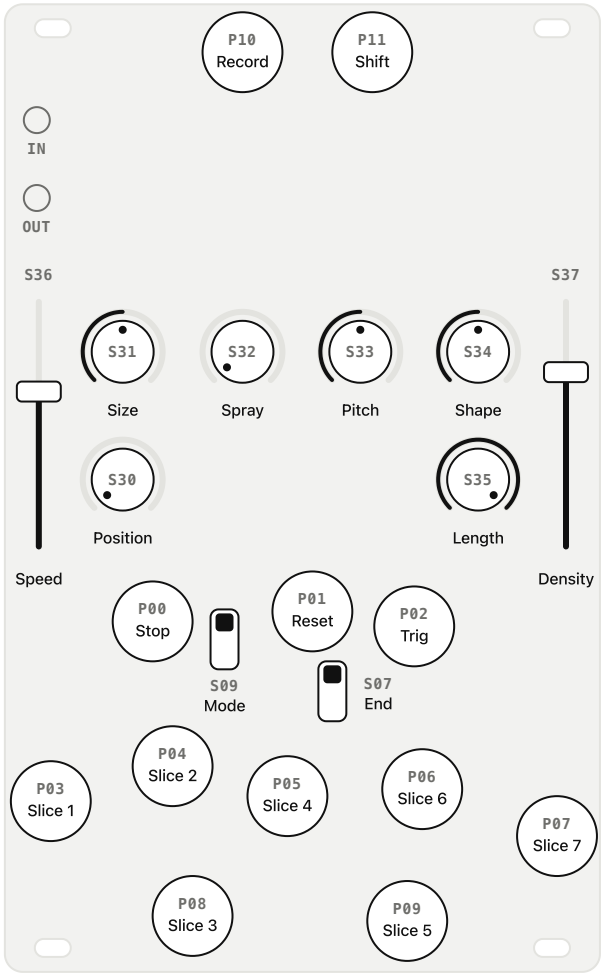
Simple Touch

Clessidra

It's the instrument where you pour grains of sound from single drops into a dense cloud.

francescodimaggio.nl/clessidra

Synthux Design Residency 2026



Record a sound, then play it as grains: Speed through it, Density how they fall.

Two layers: each knob alone, and its twin with P11 held. Nothing jumps when they change.

Density at the bottom: no grains but the slices'. S09 at its centre: frozen.

FADERS AND KNOBS

S36	**Speed**. Playback speed, 0 to 4×, the pitch kept. Frozen, it scrubs through the loop. *P11*: **Volume**. Output level, off to 0 dB.
S37	**Density**. Grains per second, 0.25 to 130, whatever their size. None at the bottom. *P11*: **Jitter**. When grains start, from a regular pulse to random, as many per second.
S30	**Position**. Where the loop starts. The playhead moves with it. *P11*: **Filter**. Low-pass below the centre, off at the centre, high-pass above.
S35	**Length**. How much of the rest the loop takes: at the top, to the end. *P11*: **Reverb**. Reverb on the grains, from none to long tails.
S31	**Size**. Grain length, 5 ms to 2 s, 100 ms at the centre. *P11*: **Random size**. Each grain a different length, up to two octaves either way.
S32	**Spray**. How far grains stray from the playhead, up to the whole loop, and how wide across the stereo field. *P11*: **Reverse**. How many grains read the other way: none, some, all. The phrase keeps its direction, each grain turns.
S33	**Pitch**. Transpose up to two octaves either way, with stops at musical intervals. *P11*: **Random pitch**. Off at the centre. Above: each grain a different pitch, up to an octave either way. Below: in octaves, more grains an octave up or down.
S34	**Shape**. Grain envelope: reversed attack, smooth at the centre, sharp attack. *P11*: **Random shape**. Each grain a different envelope, from reversed to sharp.

PADS AND SWITCHES

S09	**Mode**. Forward, freeze or reverse. Frozen, Speed scrubs through the loop.
S07	**End**. At the loop's end: once and no more, turned over to run back, or round again. Trig pours it again.
P10	**Record**. Tap to keep the last 8 s, hold to record up to 30 s. With Shift: listen.
P11	**Shift**. Hold for each knob’s second function, shown in orange.
P00	**Stop**. Stops as a tape machine, slowing to a halt. Touch again to start.
P01	**Reset**. Second functions back to their defaults.
P02	**Trig**. Playback from the loop's start, with a grain at once: back on the beat. Stopped: starts at once.
P03–P09	**Slices**. The loop in seven slices: its start, then the recording's strongest attacks inside it. Without attacks, as in a held note, even parts. Left to right on the board: P03, P04, P08, P05, P09, P06, P07. A touch: the playhead there, a grain at once, the beat from it. Held: it stays there.

TO PLAY

Rhythm	Density low, Jitter (P11 + S37) down, Shape up, Size short. Size sets how long each hit lasts
Drops	Density near the bottom, Size short: single grains, now and then
Drums by hand	Density at the bottom, Shape up: tap the slices, several at once
Cloud	Density at the top, Spray up, Size long, Reverb (P11 + S35) up
Stretch	Speed at ¼ or ½, Density up. S09 down to hear it backwards
A moment held	S09 at the centre, Density up, a little Spray. Speed scrubs the point
One pour	S07 down: the loop plays once and stops. Trig pours it again
Shimmer	Random pitch (P11 + S33) below the centre: octaves
Backwards syllables	Reverse (P11 + S32) at the top: the phrase in order, each grain backwards

LED

Glow	Grains falling, dimmer as the sand runs out

`Flashing`	Recording
`Pulse`	At a stop: unison, speed 1×, 100 ms, Hann, filter off, random pitch off
`Steady`	P11 held: the twin last turned